



Linear regression

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Agenda

Linear regression

Least squares method

Testing the significance of the coefficients α i β

Testing the significance of the whole model

Multiple linear regression





Measuring dependence

- We are often interested in whether there is a relationship between the two variables. Does one variable affect another?
- Non-parametric measurement of dependence - between two variables that are categorical - Contingency table (CrossTabs)
- Parametric measurement of the dependence - between two variables that are continuous – coefficient of correlation





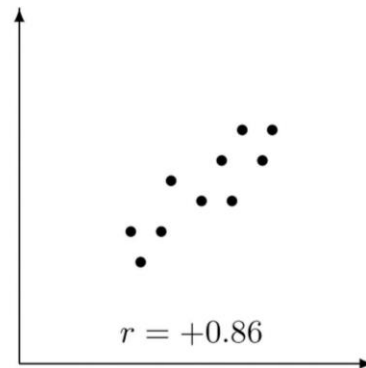
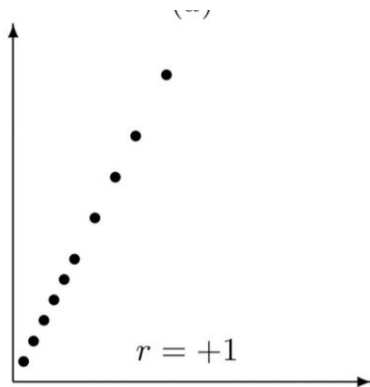
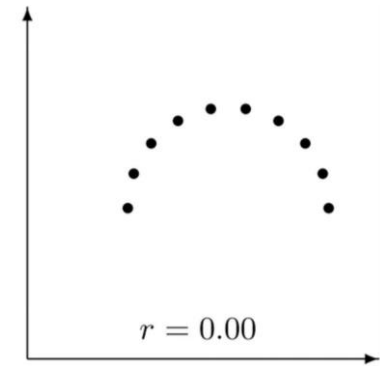
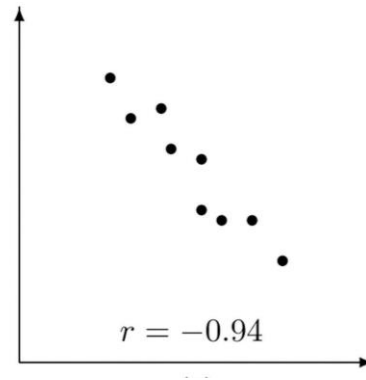
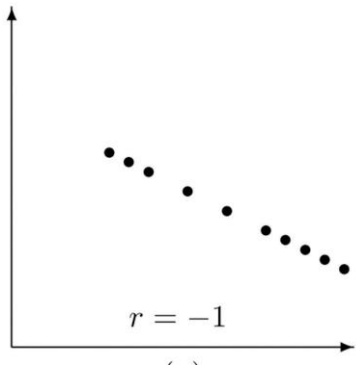
Coefficient of correlation

- A measure of the linear relationship between two continuous variables
- It gives us information about the direction of dependence and the strength of the connection between the observed variables
- Takes a value in the interval from -1 to 1





Coefficient of correlation





Linear regression

- Parametric and non-parametric measurement of dependence does not allow us to evaluate or predict the value of one phenomenon based on knowledge of the value of another phenomenon.



- IDEA: Mathematically model the dependence between the variables X and Y and use the obtained model for prediction and estimation





Relationship between the dependent and independent variable

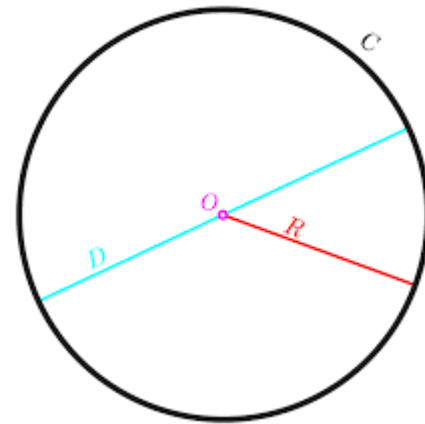
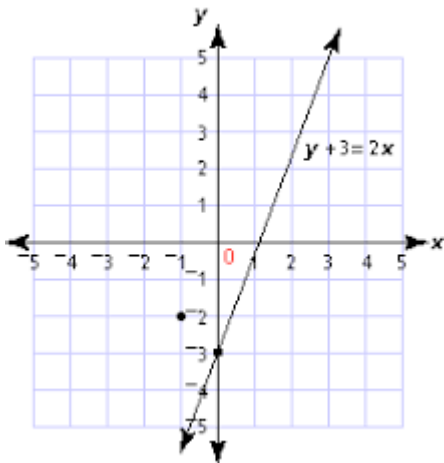
- The variables we observe are divided into:
 - Dependent - We denote it by Y and it is a variable that we want to model
 - Independent - It can be one or more of them, we mark it with X , ie $X_1, X_2, ..$
- In general, the relations that connect one dependent variable Y , with one or more independent variables X_i can be:
 - Deterministic
 - Stochastic





Relationship between the dependent and independent variable

- Deterministic
 - One value of the independent X always gives the same value of the dependent Y
 - Examples: line equation, surface area of a circle





Relationship between the dependent and independent variable

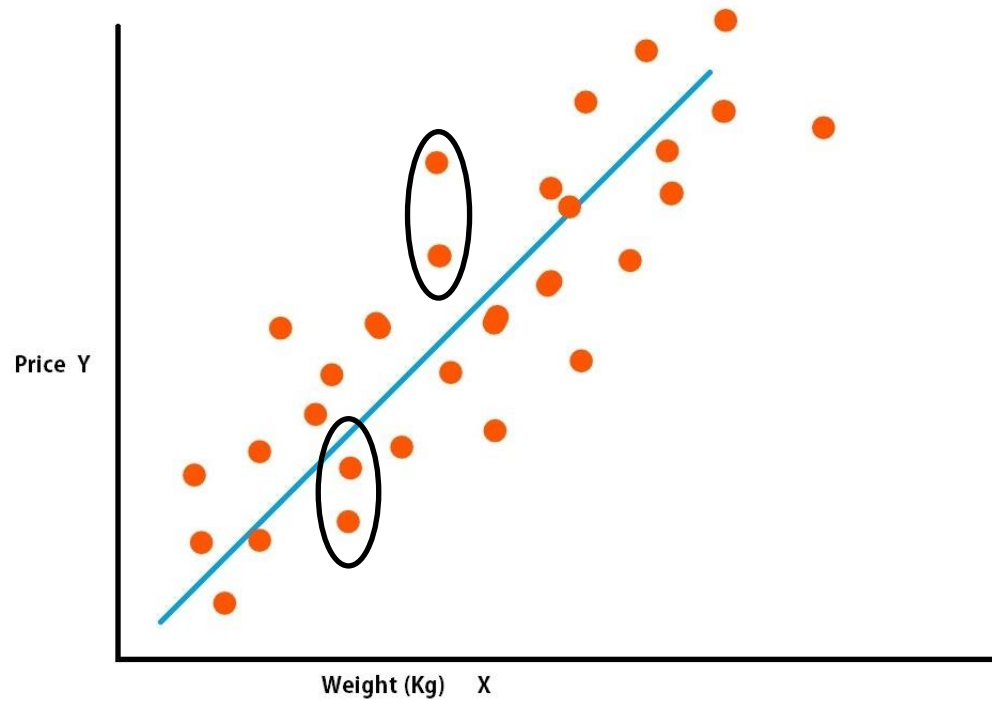
- A stochastic relationship is one in which one value of the independent variable X corresponds to a series of possible values of the dependent variable Y .
- Each of these values can be taken by the dependent variable Y with a corresponding probability, and since these values cannot be predicted with certainty, they represent a random variable.
- One value $X \rightarrow$ several different values Y





Relationship between the dependent and independent variable

- stochastic relationship:





Steps when conducting Linear regression

1. identification of variables
2. selection of a random sample
3. constructing dependency diagrams – Scatter plots
4. evaluation of model parameters
5. testing the significance of the slope and the model
6. using the model to estimate the average value of the dependent variable and predict individual values of Y_p





Steps 1 and 2

1. Identification of variables

When conducting regression analysis, the key step is to correctly conclude from the verbal description which variable is dependent Y and which is the independent X - read the task carefully

2. Selection of a random sample

We will always be given a sample in our tasks





Step 3

3. Constructing a dependency diagram

We construct the dependence diagram by plotting the values of the independent variable on the X axis, and plotting the values of the dependent variable on the Y axis.

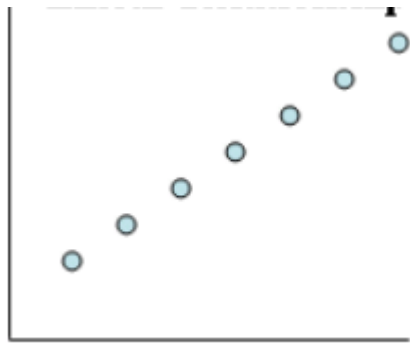
It gives us information about:

- The type of relationship - whether it is deterministic or stochastic
- Is it linear or curvilinear
- Dependency direction - whether both variables change in the same direction
- Strength





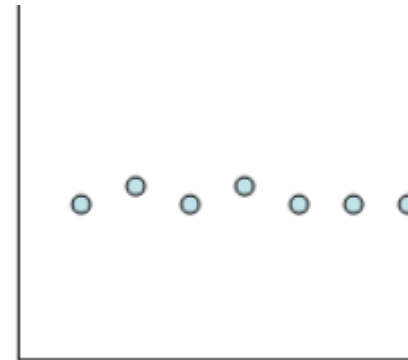
Step 3



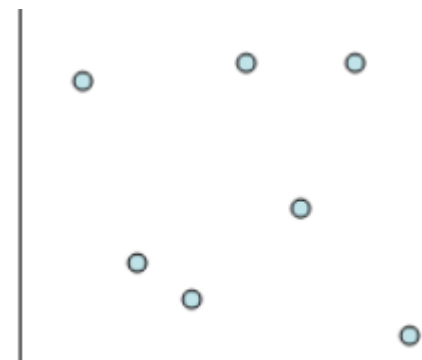
Linear
Deterministic
Direct



Linear
Deterministic
Inverse



No relationship

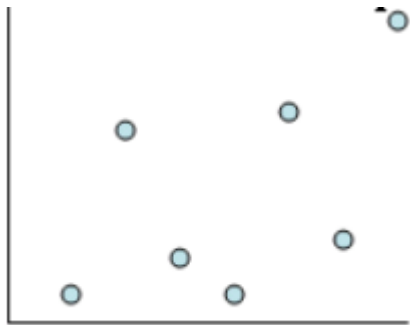


No relationship

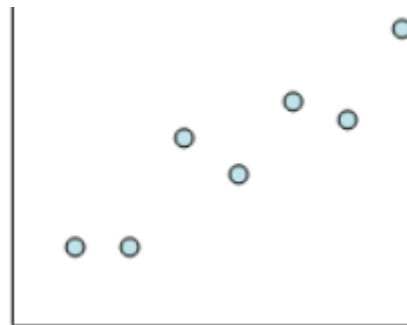




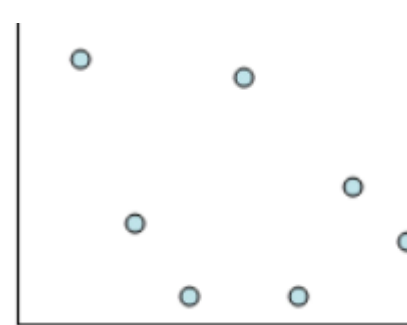
Step 3



Linear
Stochastic
Direct
Weak



Linear
Stochastic
Direct
Strong



Linear
Stochastic
Inverse
Weak



Linear
Stochastic
Inverse
Strong





Step 4

- For stochastic dependence, we have certain deviations of points from the line, and therefore, we are interested in the line that best represents the drawn points in the scatter plot.
- **Simple LRM on the population**
$$Y_i = \alpha X_i + \beta + \varepsilon_i$$
- Y_i – dependent variable
- X_i – independent variable
- α – slope
- β – constant
- ε_i – white noise (stochastic part)





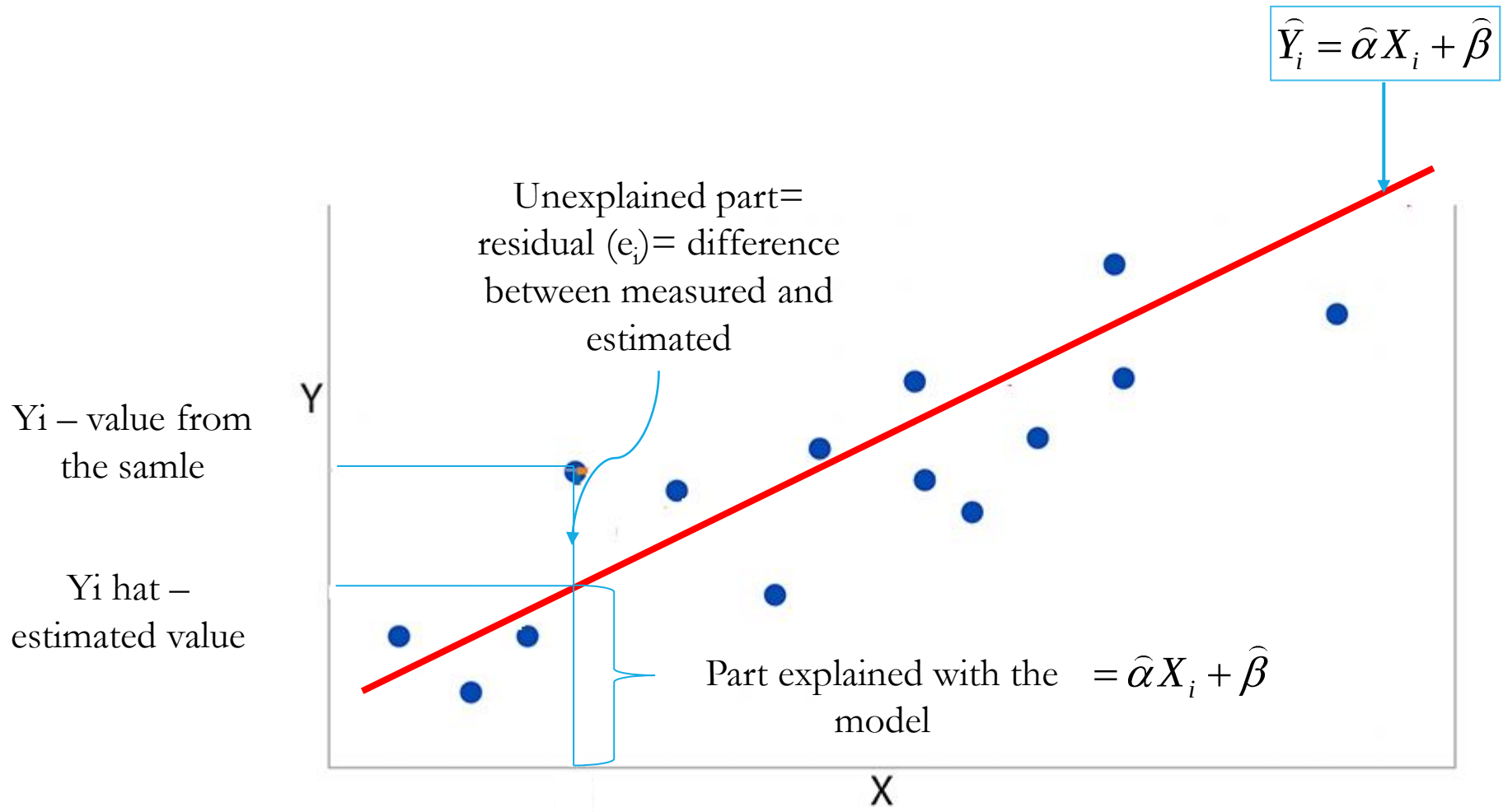
Step 4

- **Simple LRM on the sample**

$$\hat{Y}_i = \hat{\alpha}X_i + \hat{\beta}$$

- $\hat{Y}_i$ – estimated value of the dependent variable
- X_i – independent variable
- $\hat{\alpha}$ – estimated slope
- $\hat{\beta}$ – estimated constant







Least squares method

- We define the deviations of empirical points (obtained values) from the regression line. These deviations are vertical deviations between the actual and estimated value of y and are called residuals.

$$e_i = Y_i - \hat{Y}_i = Y_i - (\hat{\alpha} X_i + \hat{\beta})$$

- **Idea:** Draw a line through the points, ie estimate α and β , so that the deviations between the measured values of Y_i and the estimated values of Y_i are as small as possible
- Minimal + Squared





Least squares method

- Obtained estimates:

$$\hat{\alpha} = \frac{\frac{1}{n} \sum X_i Y_i - \bar{x}\bar{y}}{S_x^2}$$

$$\hat{\beta} = \bar{y} - \hat{\alpha}\bar{x}$$





Residual variance

- Variance of the regression

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{k=1}^n (Y_i - \hat{Y}_i)^2 = \frac{1}{n} \left(\sum Y_i^2 - \hat{\alpha} \sum Y_i X_i - \hat{\beta} \sum Y_i \right)$$

- Standard error of the regression – square root of the variance of regression

$$\hat{\sigma} = \sqrt{\hat{\sigma}^2}$$

- Tell us how much our data deviates from the regression line
- The smaller, the better





Step 5

- We can determine the scores α and β on each sample
- However, the scores obtained do not always have to be statistically significant
- We can individually examine the significance of the coefficients α and β
- We can individually examine the significance of the whole model





Significance of the coefficients α and β

- Testing the significance of the coefficient α
- $H_0(\alpha = 0)$ The coefficient is not statistically significant, X does not affect Y
- $H_1(\alpha \neq 0)$ The coefficient is statistically significant, X affects Y
- If the test leads us to the conclusion that $\alpha = 0$, we must not use the estimated regression line for the purpose of prediction.





Significance of the coefficients α and β

- Testing the significance of the coefficient β
- $H_0(\beta = 0)$ The constant is not statistically significant in the model
- $H_1(\beta \neq 0)$ The constant is statistically significant in the model





Significance of the whole model

- Testing the significance of the whole model
- H_0 The model is not statistically significant
- H_1 The model is statistically significant





Linear regression – Example 1

Examples on the data set **IDIran**.sav :

1. Using scatter plot describe the relationship between Mobile-cellular subscriptions in 2011 and 2012
2. Using linear regression model the Mobile-cellular subscriptions in 2012 using Mobile-cellular subscriptions in 2011. Is the model statistically significant?
3. Which proportion of Mobile-cellular subscriptions in 2012 can be explained using Mobile-cellular subscriptions in 2011?





Example 1

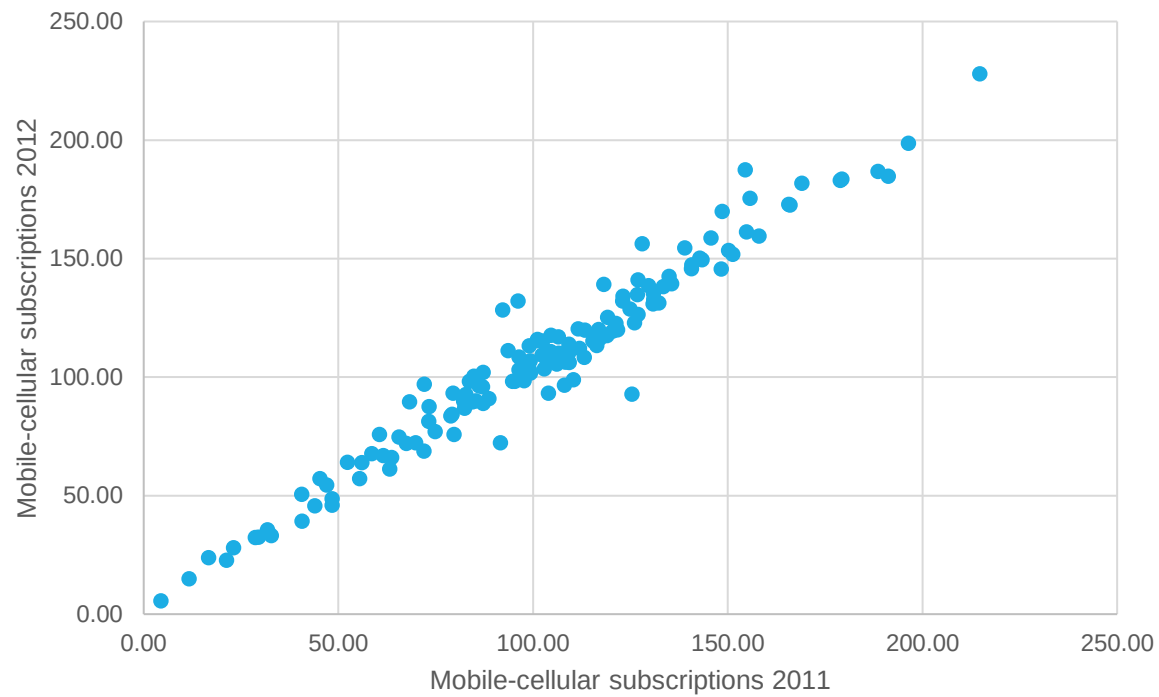
We create the scatter plot

- Graphs – Legacy dialogs – Scatter/Dot
- Simple scatter – Define
- Y axis – Dependent, Y – *Mobile 2012*
- X axis – Independent, X – *Mobile 2011*
- Ok 😊





Example 1





Example 1

Examples:

1. Using scatter plot describe the relationship between Mobile-cellular subscriptions in 2011 and 2012
 - The relationship is **strong, direct, and linear**





Example 1

- We will obtain many results with one analysis
- Analyze – Regression – Linear
- Dependent – Mobile 2012
- Independent(s) – Mobile 2011
- The Statistics section we will check Descriptives i Confidence intervals
- Ok 😊





Example 1

- Descriptive statistics – Table *Descriptives*
 - Mean Y – Mobile 2012 = 107.739
 - Mean X – Mobile 2011 = 102.352
- Coefficient of correlation – Table *Correlations*
 - Pearson Correlation - $r = 0.976$





Example 1

Examples:

2. Using linear regression model the Mobile-cellular subscriptions in 2012 using Mobile-cellular subscriptions in 2011. Is the model statistically significant?

In this case, the values of $\mathbf{b}_0$ and $\mathbf{b}_1$ are as follows: $\mathbf{b}_0 = 5.324$, $\mathbf{b}_1 = 1.001$,

$$\hat{Y}_i = 5.324 + 1.001X_i$$

Both the coefficient and the model are statistically significant as p values are less than 0.05





Example 1

Standard error of the regression and coefficient of determination—
Table *Model summary*

column Std. error of the estimate — $\sigma_e = 8.812$

column R square — $r^2 = 0.952$

R square: 95.2% variability of Mobile 2012 can be explained with
Mobile 2011





Linear regression

Examples:

3. Which proportion of Mobile-cellular subscriptions in 2012 can be explained using Mobile-cellular subscriptions in 2011?

The coefficient of determination is 0.9522, therefore 95.22% of variability of Mobile-cellular subscriptions in 2012 can be explained with Mobile-cellular subscriptions in 2011





Forecasting

The basic idea of LRM is that it is used to predict and evaluate Y based on X values

Can every LRM be used for this purpose? – **NO**

In order to correctly use LRM for forecasting, it is necessary that:

- The regression line represents the empirical data well ($r^2 > 0.5$)
- There is a linear relationship between X and Y , ie. that the t-test showed ($\beta \neq 0$)





Forecasting – Estimation and prediction

For some selected value of the variable X , which we denote by X_p , it is possible to:

- **Estimate the average value $E(Y_p)$** – we estimate the average value Y for all entities that have a given X_p
- **Predict the individual value of Y_p** - we predict the value of Y for a separate, specific, defined entity that has a given X_p





Forecasting – Estimation and prediction

If we are evaluating and predicting for the same X_p , are the intervals the same length? – **NO**

The difference is that $E(Y_p)$ represents a constant while Y_p is a random variable because it depends on the random error.

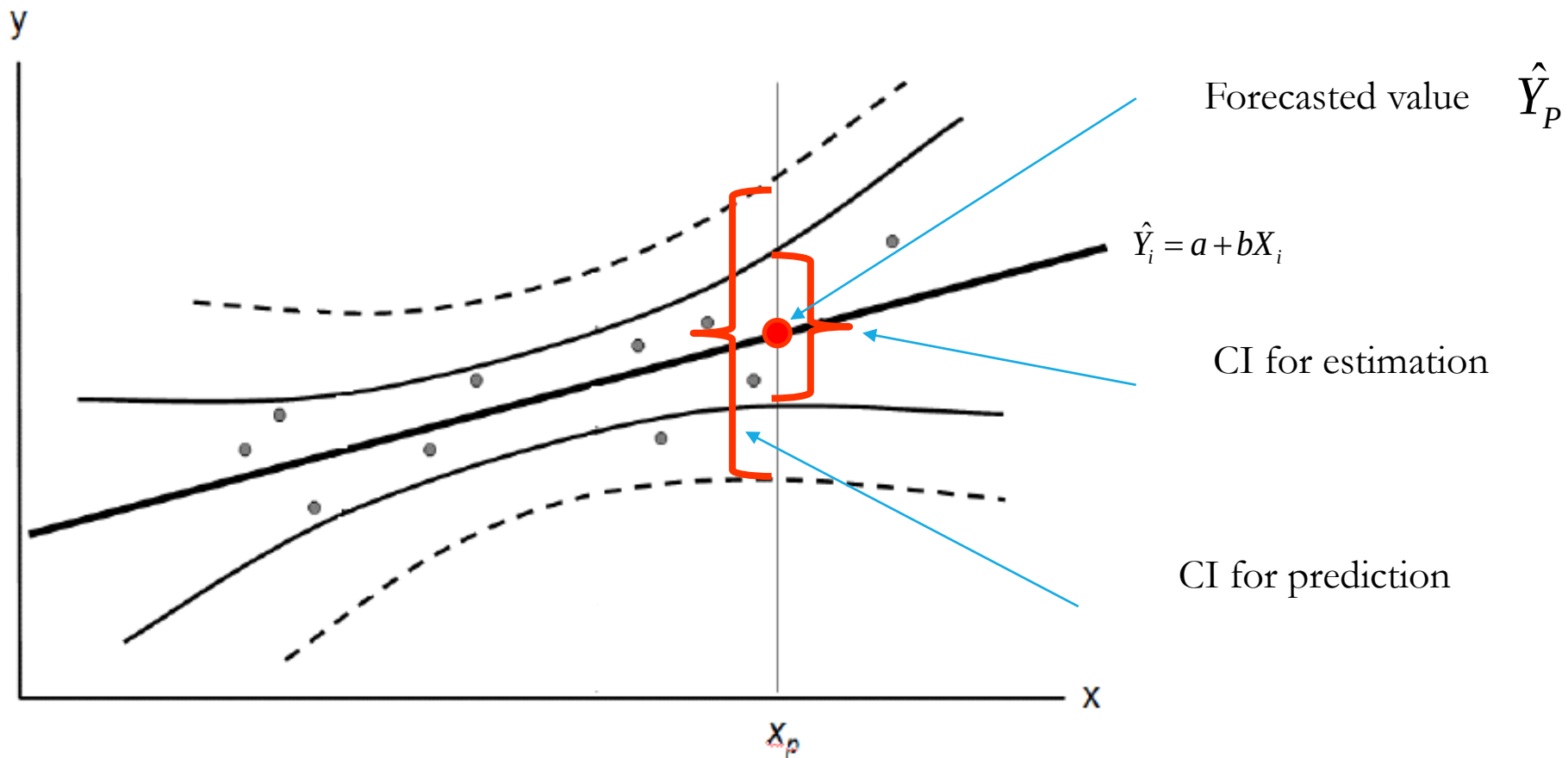
Therefore, the prediction of individual values is more uncertain, so the corresponding prediction interval will be wider than the interval estimate $E(Y_p)$.

The confidence interval for the individual prediction is wider because we take random error into account





Forecasting – Estimation and prediction





Forecasting – Estimation and prediction

- Whether we are estimating or predicting, when we include X_p in the evaluated model, we get an estimated $\hat{Y}_p$:

$$\hat{Y}_p = a + bX_p$$

- The point estimate when estimating or predicting is the same and amounts $\hat{Y}_p$
- However, the confidence intervals around that point estimate differ depending on whether we are estimating or predicting





Linear regression – Example 1

Examples on the data set **IDIrak.sav** :

Observe the model of the Mobile-cellular subscriptions in 2012 using Mobile-cellular subscriptions in 2011.

1. Predict the value for all countries that have Mobile-cellular subscriptions in 2011 45.4
2. Evaluate the value for Serbia who has Mobile-cellular subscriptions in 2011 57.2





Linear regression – Example 1

- As we showed before, the model is statistically significant and we can use it for estimation and prediction
- **IMPORTANT** - Given that we need to perform estimation and prediction for new values which might or might not be in the database, for the independent variable, Mobile-cellular subscriptions in 2011, we have to enter the two new values – 45.4 and 57.2
- The dataset is ready for further work





Linear regression – Example 1

File Edit View Data Transform Analyze Graphs Utilities Extensions Window Help

147 : Mobile2011

	Country	Fixed200 8	Mobile20 08	Literacy2 008	Fixed201 0	Mobile20 10	Literacy2 010	Fixed201 1	Mobile20 11	Literacy2 011	Fixed
130	Trinidad & Tobago	23.70	135.70	98.70	21.90	141.20	98.70	21.70	135.60	98.80	
131	Tunisia	12.10	83.90	77.60	12.30	106.00	77.60	11.50	116.90	79.10	
132	Turkey	24.70	92.80	88.70	22.30	84.90	90.80	20.70	88.70	94.10	
133	Uganda	.50	27.30	71.40	1.00	38.40	71.40	1.30	48.40	73.20	
134	Ukraine	28.70	121.10	99.70	28.50	118.70	99.70	28.10	123.00	99.70	
135	United Arab Emirates	25.70	150.80	90.00	19.70	145.50	90.00	23.10	148.60	90.00	
136	United Kingdom	55.90	125.20	99.00	53.70	130.20	99.00	53.30	130.80	99.00	
137	United States	53.40	85.70	99.00	48.70	89.90	99.00	45.80	95.30	99.00	
138	Uruguay	28.70	104.80	98.20	28.60	131.70	98.30	28.50	140.80	98.10	
139	Uzbekistan	6.90	46.20	99.30	6.80	76.30	99.30	6.90	91.60	99.40	
140	Venezuela	22.90	97.70	95.20	24.40	96.20	95.20	24.90	97.80	95.50	
141	Viet Nam	17.20	87.10	92.80	18.70	175.30	92.80	11.50	143.40	93.40	
142	Yemen	4.20	28.50	62.40	4.30	46.10	62.40	4.30	47.00	65.30	
143	Zambia	.70	28.60	70.90	.70	37.80	70.90	.60	60.60	61.40	
144	Zimbabwe	2.80	13.30	91.90	3.00	59.70	91.90	2.80	72.10	83.60	
145		-	-	-	-	-	-	-	45.40	-	
146		-	-	-	-	-	-	-	57.20	-	
147											
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1

Data View Variable View





Linear regression – Example 1

- We will get the results for all other requests with one analysis run
- Analyze – Regression – Linear
- Dependent - Dependent - Calories
- Independent(s) - Fats
- In the Statistics section, check Descriptives and Confidence interval -
We want basic descriptive statistics of variables and a confidence interval for β
- We've done these requests so far, however we need to specify a command to estimate and predict





Linear regression – Example 1

- We will get the results for all other requests with one analysis run
- We are going to part of the Save
- In the Predicted values section - we check Unstandardized - we want SPSS to save the unstandardized predicted value
- In the section Prediction intervals - we check Mean (IP for evaluation) and Individual (IP of predicted individual values)
- Continue
- Ok 😊





Linear regression – Example 1

Linear Regression: Save

Predicted Values

- ☒ Unstandardized
- ☐ Standardized
- ☐ Adjusted
- ☐ S.E. of mean predictions

Residuals

- ☐ Unstandardized
- ☐ Standardized
- ☐ Studentized
- ☐ Deleted
- ☐ Studentized deleted

Distances

- ☐ Mahalanobis
- ☐ Cook's
- ☐ Leverage values

Prediction Intervals

- ☒ Mean ☒ Individual
- Confidence Interval: %

Influence Statistics

- ☐ DfBeta(s)
- ☐ Standardized DfBeta(s)
- ☐ DfFit
- ☐ Standardized DfFit
- ☐ Covariance ratio

Coefficient statistics

- ☐ Create coefficient statistics
- ☒ Create a new dataset
Dataset name:
- ☒ Write a new data file

Export model information to XML file

☒ Include the covariance matrix





Linear regression – Example 1

- **Predict the value for all countries that have Mobile-cellular subscriptions in 2011 45.4**
- We read the evaluation results from the Data View, not from the Output window
- 5 new variables were created in the database:
- PRE_1 – Predicted value – The value that the model proposes if we insert the given X_i into it –
- LMCI_1 } Lower (Upper) mean confidence interval – Interval poverenja
- UMCI_1 } ocene prosečne vrednosti
- LICI_1 } Lower (Upper) individual confidence interval – Interval poverenja
- UICI_1 } predvidene individualne vrednosti





Linear regression – Example 1

File Edit View Data Transform Analyze Graphs Utilities Extensions Window Help																			
145 : Country																			
	Mobile2011	Ltra	Fxe	RFX	Mbil	Rlo	Ltra	Fte	H	H	C	C	C	PRE_1	LMCI_1	UMCI_1	LICI_1	UICI_1	var
128	107.20	97.40	19.70	63.0	108.2	74.0	97.40	56.5	3	2	1	2	2	112.59051	111.12762	114.05340	95.10840	130.07262	
129	111.60	93.50	9.10	95.0	120.3	45.0	93.50	76.5	2	2	1	2	2	116.99324	115.50131	118.48518	99.50868	134.47781	
130	135.60	98.80	21.20	55.5	139.4	27.0	98.80	45.0	2	2	2	2	2	141.00815	139.10106	142.91525	123.48328	158.53303	
131	116.90	79.10	10.30	89.0	120.0	46.0	79.10	106.0	2	2	1	2	2	122.29654	120.74723	123.84585	104.80698	139.78609	
132	88.70	94.10	18.60	66.0	90.80	103.0	94.10	72.5	2	2	1	1	1	94.07902	92.54103	95.61701	76.59046	111.56758	
133	48.40	73.20	.90	129.5	45.90	133.0	73.20	112.0	4	4	1	1	1	53.75399	51.27709	56.23088	36.15799	71.34998	
134	123.00	99.70	27.10	44.0	132.1	35.0	99.70	7.0	2	2	2	2	2	128.40033	126.75794	130.04271	110.90228	145.89837	
135	148.60	90.00	24.30	48.5	169.9	12.0	90.00	89.0	1	2	2	2	2	154.01623	151.76525	156.26721	136.45061	171.58185	
136	130.80	99.00	52.60	8.0	130.8	38.0	99.00	29.0	1	1	2	2	2	136.20517	134.40871	138.00163	118.69199	153.71835	
137	95.30	99.00	44.00	18.0	98.20	92.0	99.00	29.0	1	2	1	1	1	100.68312	99.20788	102.15836	83.19997	118.16627	
138	140.80	98.10	29.80	40.5	147.3	22.0	98.10	50.0	2	2	2	2	2	146.21138	144.17353	148.24924	128.67180	163.75097	
139	91.60	99.40	7.00	103.0	72.20	118.5	99.40	14.0	3	2	1	1	1	96.98082	95.47500	98.48664	79.49506	114.46658	
140	97.80	95.50	25.60	47.0	102.1	86.0	95.50	64.0	2	2	1	1	2	103.18467	101.72310	104.64625	85.70267	120.66667	
141	143.40	93.40	11.40	85.0	149.4	21.0	93.40	78.0	3	2	1	2	2	148.81300	146.70614	150.91985	131.26526	166.36073	
142	47.00	65.30	4.30	108.0	54.40	130.0	65.30	125.0	4	3	1	1	1	52.35312	49.83384	54.87239	34.75110	69.95513	
143	60.60	61.40	.60	135.5	75.80	115.0	61.40	128.0	4	4	1	1	1	65.96156	63.83565	68.08748	48.41153	83.51160	
144	72.40	93.60	2.30	119.0	96.90	95.0	83.60	105.0	4	3	1	1	1	77.48071	75.83192	79.30550	59.95134	94.90807	
145	45.40	-	-	-	-	-	-	-	-	-	-	-	-	50.75212	48.18398	53.32026	33.14305	68.36120	
146	57.20	-	-	-	-	-	-	-	-	-	-	-	-	62.55945	60.33947	64.77944	44.99777	80.12113	
147																			
148																			
149																			





Linear regression – Example 1

- **Predict the value for all countries that have Mobile-cellular subscriptions in 2011 45.4**

$$X_p = 45.40$$

$$E(Y_p) = \hat{Y}_i = 50.75$$

We read from PRE_1

$$48.184 \leq E(Y_p) \leq 53.320$$

We read from LMCI_1 and UMCI_1

Interpretation: For all countries that have Mobile2011 equal 45.4, we **estimate** they will have the **average** Mobile2012 of 50.75 with the CI from 48.184 to 53.320





Linear regression – Example 1

- Evaluate the value for Serbia who has Mobile-cellular subscriptions in 2011 57.2

$$X_p = 57.20$$

$$E(Y_p) = \hat{Y}_i = 62.56$$

We read from PRE_1

$$44.997 \leq E(Y_p) \leq 80.121$$

We read from LMCI_1 and UMCI_1

Interpretation: For Serbia who has Mobile2011 equal 57.2, we **predict** it will have the Mobile2012 of 62.56 with the CI from 44.997 to 80.121





Multiple linear regression model

- In the economy, it is normal that the behavior of a phenomenon is influenced by a number of other factors.
- For example: sales of a product are affected by:
 - Product prices
 - Income of the population
 - Advertising expenses
 - Competitive product prices
- A multiple linear regression model is an extension of a simple LRM, where in addition to the dependent variable Y , two or more independent variables are included in the model.





Multiple linear regression model

- Multiple linear regression model – on the population

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_k X_{ki} + \varepsilon_i$$

Y_i – dependent variable

$X_{1i}, X_{2i}, \dots, X_{ki}$ – independent variables

β_0 - constant

$\beta_1, \beta_2, \dots, \beta_k$ – slopes

ε_i – white noise

k – number of independent variables





MLRM with two independent variables

- MLRM with two independent variables – on the population

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \varepsilon_i$$

Y_i – dependent variable

X_{1i}, X_{2i} – independent variables

β_0 – constant

β_1 – shows the average change of the dependent variable Y when the independent variable X_1 increases by one of its units, provided that X_2 remains constant

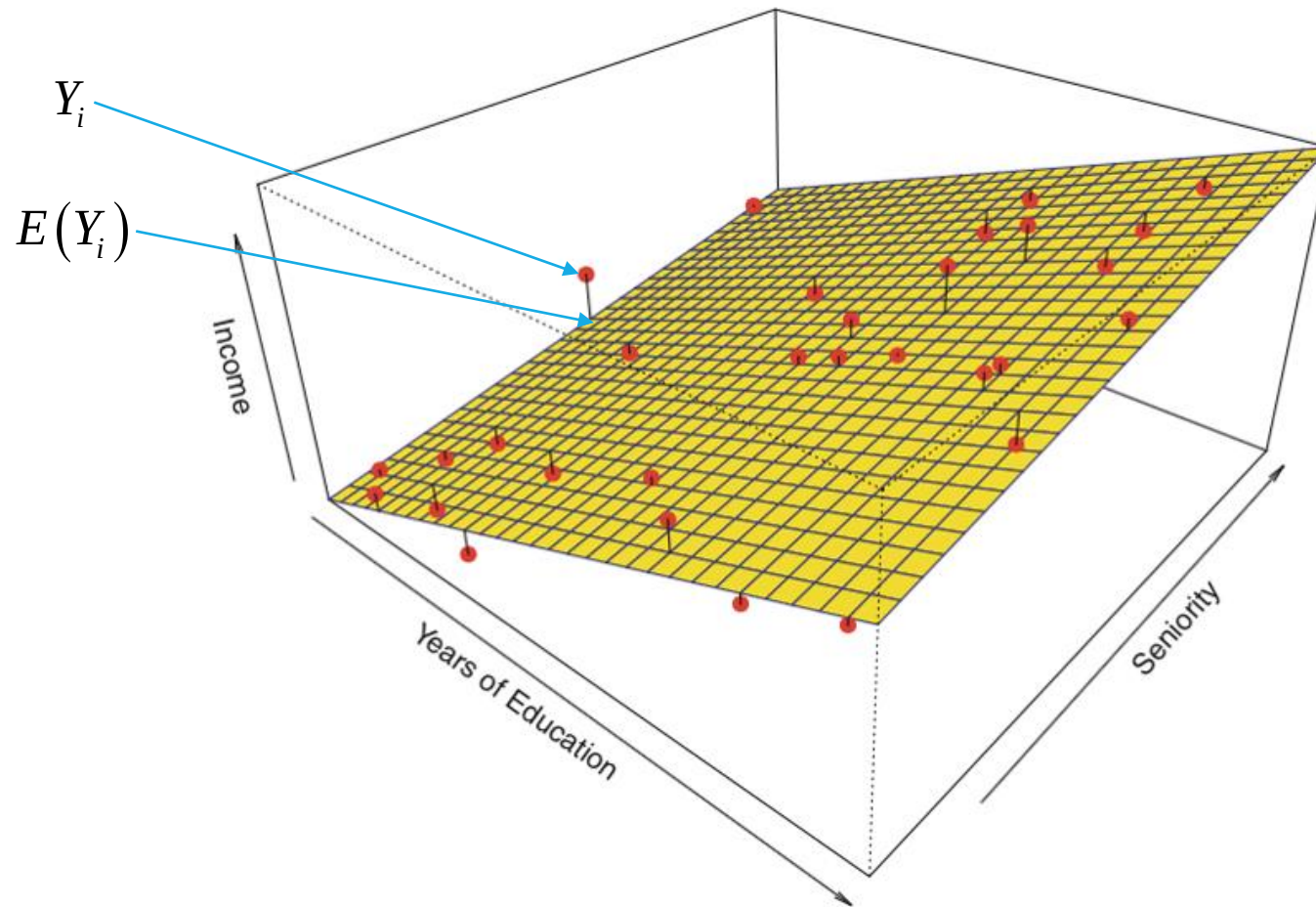
β_2 – shows the average change of the dependent variable Y when the independent variable X_2 increases by one of its units, provided that X_1 remains constant

ε_i – white noise





MLRM with two independent variables





MLRM with two independent variables

- MLRM with two independent variables – on the sample

$$\hat{Y}_i = b_0 + b_1 X_{1i} + b_2 X_{2i}$$

$\hat{Y}_i$ – estimated value of the dependent variable Y_i

X_{1i}, X_{2i} – independent variables

b_0 – estimate of β_0

b_1 – estimate of β_1

b_2 – estimate of β_2





MLRM with two independent variables

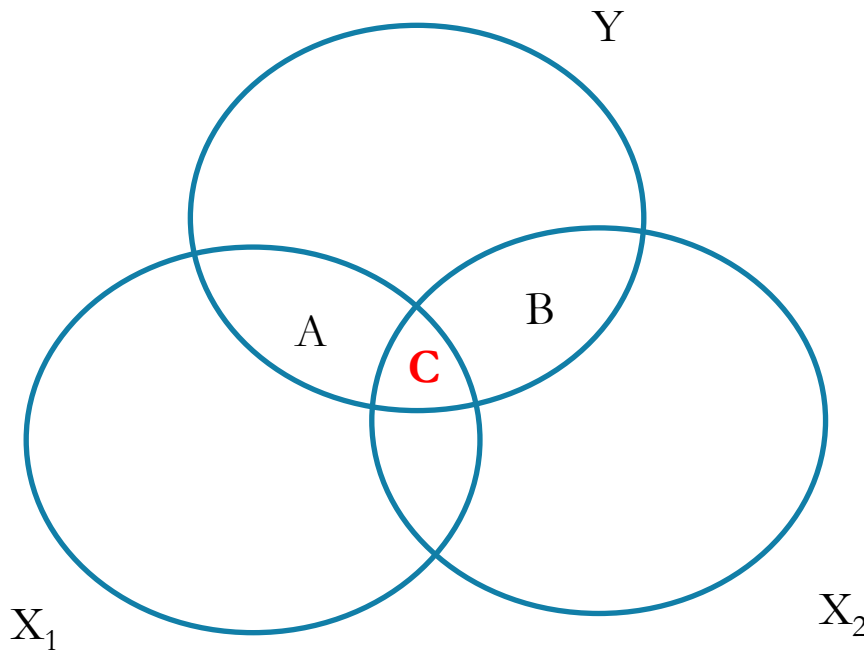
- The coefficient of multiple determination is NOT a reliable measure of model quality because:
 - It depends on the number of variables in the model and the sample size (if the number of data in the sample is small and there is a large number of independent variables, the value of R^2 will be close to 1 even though they individually do not affect the dependent variable much).
 - By including a new independent variable in the model, R^2 increases regardless of the actual impact.





MLRM with two independent variables

- Coefficient of multiple determination



Surface C - the part where X_1 and X_2 overlap when explaining Y
- the amount of explained variability is doubled and "artificially" increased

Solution: Eliminate duplicity





MLRM with two independent variables

- Adjusted R square
- represents the corrected coefficient of determination in order to obtain an indicator that will not increase with the increase in the number of explanatory variables.
- shows the share of the explained variability in the total variability - expressed as a percentage and interpreted as % of the variability Y explained through X_{1i} and X_{2i}
- when interpreting the output we will interpret ONLY the adjusted coefficient of multiple determination





MLRM with two independent variables

- We can make a regression model on every sample, but are the coefficients β_1 and β_2 statistically significant and is the whole model statistically significant?
- t-test - Is the coefficient β_1 / β_2 statistically significant?

$$H_0(\beta_1 = 0)$$

$$H_1(\beta_1 \neq 0)$$

$$H_0(\beta_2 = 0)$$

$$H_1(\beta_2 \neq 0)$$

$$t_1 = \frac{b_1}{\sqrt{V(b_1)}} : t_{N-k-1}$$

$$t_2 = \frac{b_2}{\sqrt{V(b_2)}} : t_{N-k-1}$$





MLRM with two independent variables

- We can make a regression model on every sample, but are the coefficients β_1 and β_2 statistically significant and is the whole model statistically significant?
- F-test – Is the whole model statistically significant?
- H_0 : The model is not statistically significant
- H_1 : The model is statistically significant





MLRM with two independent variables

- It can happen that, for example, the coefficient β_1 is statistically significant, and the coefficient β_2 is not, and that the whole MLRM is statistically significant - HOW?
- the t-test observes the individual influence of independent variables on the dependent one, and the F-test observes their joint influence. So, one independent variable may not individually have a statistically significant effect on the dependent one, but together with another independent one, it forms a model that is statistically significant.





Linear regression

Examples on the data set IDIrak.sav :

1. Model and explore how Fixed2008 and Mobile2008 impact Literacy2008





Example 2

- We will obtain many results with one analysis
- Analyze – Regression – Linear
- Dependent – Literacy2008
- Independent(s) – Mobile2008, Fixed2008
- The Statistics section we will check Descriptives i Confidence intervals
- Ok 😊





Example 2

- Descriptive statistics – Table *Descriptives*
 - Mean Y – Literacy2008 = 85.77
 - Mean X1 – Fixed2008 = 21.28
 - Mean X2 – Mobile2008 = 81.50





Example 2

In this case, the values of $\mathbf{b}_0$, $\mathbf{b}_1$ and $\mathbf{b}_2$ are as follows: $\mathbf{b}_0 = 62.226$, $\mathbf{b}_1 = 0.353$, $\mathbf{b}_2 = 0.197$,

$$\hat{Y}_i = 62.226 + 0.353\text{Fixed}X_i + 0.197\text{Mobile}X_i$$

Both the coefficients and the model are statistically significant as p values are less than 0.05





Example 2

Standard error of the regression and coefficient of determination—
Table *Model summary*

column Std. error of the estimate — $\sigma_e = 12.80$

column R square — $R^2 = 0.515$

column Adjusted R square - $\bar{R}^2 = 0.508$

R square: 50.8% variability of Literacy2008 can be explained with
Mobile2008 and Fixed2008





Linear regression

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